higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE (VOCATIONAL)

ELECTRICAL PRINCIPLES AND PRACTICE
NQF LEVEL 2

27 FEBRUARY 2019

This marking guideline consists of 7 pages.

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QUESTION 1

1.1 Time is the quantity used to measure the duration of events, compare intervals of events and to quantify a rate of change like the movement of an object. (2)

1.2 $2{,}3\ \text{cm}^3 = 2{,}3 \times (10^{-2})^3$ ✓
$= 2{,}3 \times 10^{-6}\ m^3$ ✓
$= 0{,}0000023\ m^3$ (2)

1.3
- Symbols for physical quantities are written in italics.
- Write the value of a quantity as a number followed by a space and then the unit symbol.
- Spaces are used to separate thousands.
- Use a decimal comma rather than a decimal point. (Any 3 × 1) (3)

1.4 One newton is the amount of force that imparts an acceleration of 1 m/s^2 to a mass of 1 kg.

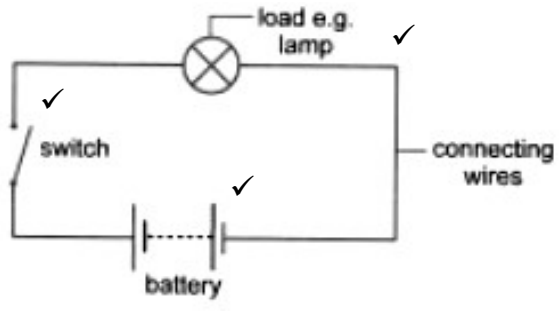
$F = m \times a$
$= \text{kg} \times \text{m/s}^2$ ✓
$= N$ ✓ (3)

[10]

QUESTION 2

2.1 Electrons are particles with a negative charge that orbit the nucleus of an atom,✓✓ whilst free electrons are not attached to the nucleus of an atom and free to move when external energy is applied.✓✓ (2 + 2) (4)

2.2



(3)

2.3 Ohm's law states that the current flowing in a circuit is directly proportional to the applied voltage and inversely proportional to the resistance, provided that the temperature of the resistance remains constant.

Mathematical expression: $I = \frac{V}{R}$ (3)

2.4
- The generation of AC is cheaper than DC.
- AC voltages can easily be stepped up or down with a transformer for transmission purposes.
- AC voltage can be transmitted to distant places with little loss in AC power.
- AC can easily be converted into DC by using rectifier circuits.
- AC generators have a larger capacity than DC generators.
- AC motors are less expensive to build and require less maintenance than DC motors. (Any 3 × 1) (3)

2.5
- Length of the conductor
- Cross-sectional area of the conductor
- Resistivity or type of material
- Temperature (Any 3 × 1) (3)

2.6 2.6.1

$$I = \frac{P}{V}$$
$$= \frac{4\ 000}{250}\ \checkmark$$
$$= 16\ A\ \checkmark$$

2.6.2

$$R = \frac{V^2}{P}$$
$$¿ \frac{250^2}{4\ 000}\ \checkmark$$
$$¿ 15{,}625\ \Omega\ \checkmark$$

Alternative:

$$R = \frac{V}{I}$$
$$= \frac{250}{16}$$
$$= 15{,}625\ \Omega$$

2.6.3

$$Q = I^2 R.t$$
$$¿ 16^2 \times 15{,}625 \times 60 \times 60\ \checkmark$$
$$¿ 14{,}4\ \text{MJ}\ \checkmark$$

Alternative:

$$Q = \frac{V^2}{R}\ t$$
$$= \frac{250^2}{15{,}625} \times (60 \times 60)$$
$$= 14{,}4\ \text{MJ}$$

(3 × 2) (6)

2.7 Place your right hand over the coil with the fingers pointing in the direction of the current✓ and with your thumb at 90° to your fingers.✓ Your thumb will point in the direction of the north.✓ (3)

[25]

QUESTION 3

3.1 3.1.1 $E_T = e_1 = 1{,}5\ V$ (1)

3.1.2
$$r_T = \frac{r_1}{n}$$
$$= \frac{0{,}3}{6}$$
$$= 0{,}05\ \Omega$$ (1)

3.1.3
$$I = \frac{E_T}{R + r_T}$$ ✓
$$= \frac{1{,}5}{2{,}45 + 0{,}05}$$ ✓
$$= 0{,}6\ A$$ ✓ (3)

3.1.4
$$\text{Pd} = I_T R_T$$
$$= 0{,}6 \times 2{,}45$$ ✓
$$= 1{,}47\ V$$ ✓ (2)

3.2 In a step-up transformer the secondary coil is bigger with more windings than the primary coil,✓✓ while in a step-down transformer the secondary coil is smaller with fewer windings than the primary coil.✓✓ (2 + 2) (4)

3.3
$$\text{Heaters} = 4 \times 1\,000 = 4\,000$$ ✓
$$\text{Floodlights} = 3 \times 100 = 300$$ ✓
$$\text{Luminaires} = 5 \times 100 = 500$$ ✓

$$\text{Total load per phase} = \frac{4\,800}{3}$$ ✓
$$= 1\,600\ W$$ ✓ (5)

3.4 3.4.1
$$\frac{N_1}{N_2}$$ ✓
$$\frac{200}{60}$$ ✓
$$3{,}333:1$$ ✓

3.4.2 $$\frac{N_1}{N_2}=\frac{V_1}{V_2}$$

$$\frac{200}{60}=\frac{230}{V_2}$$

$$V_2=\frac{230\times 60}{200}$$

$$\therefore 69\ V$$

3.4.3 $$\frac{N_1}{N_2}=\frac{I_2}{I_1}$$

$$\frac{200}{60}=\frac{I_2}{6}$$

$$I_2=\frac{6\times 200}{60}$$

$$\therefore 20\ A$$

(3 × 3) (9)
[25]

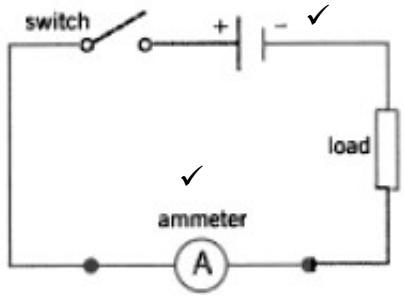
QUESTION 4

4.1 4.1.1 Parallel
4.1.2 Parallel
4.1.3 Series

(3 × 1) (3)

4.2 When current flows through the coil, the magnetic field set up by it interacts with the field✓ produced by the permanent magnet.✓ This causes the coil to pivot.✓ (3)

4.3



(2)

4.4
- Before taking any measurement, make sure that the pointer is at zero if it is an analogue meter.
- Do not clamp it on a bare conductor.
- Always start with the highest scale on the meter when measuring unknown values. (3)

4.5 4.5.1 Series resistor or multiplier✓ connected in series with the voltmeter✓

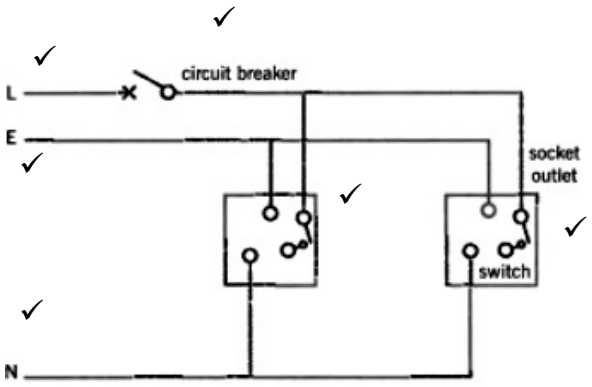
4.5.2 Shunt resistor✓ connected in parallel with the ammeter✓

(2 × 2) (4)
[15]

QUESTION 5

5.1
- Use the correct symbols.
- Use a suitable symbol orientation.
- Pay attention to the arrangement of symbols on the diagram.
- Pay attention to the routing of interconnections.
- Make sure the drawing is neat. (Any 3 × 1) (3)

5.2



(6)

5.3
- 1 × 10 A single-phase circuit breaker
- 1 × 20 A single-phase circuit breaker
- 1,5 mm^2 red wire
- 1,5 mm^2 black wire
- 1,5 mm^2 green wire
- 2,5 mm^2 red wire
- 2,5 mm^2 black wire
- 2,5 mm^2 green or bare wire
- 3 × one-lever one-way switch
- 3 × lamp holders
- 3 × single-pole plugs (Any 6 × 1) (6)

[15]

QUESTION 6

6.1 6.1.1 Electrical conductors allow electric current to pass through them,✓ whereas insulators block the flow of electricity.✓ (2)

6.1.2
- Nonmetal with a relatively high resistance
- Able to conduct electric current (2)

6.1.3
- Synthetic material
- Hard material with a safe working temperature of 200 °C
- Brown colour (Any 2 × 1) (2)

6.2
- Under normal circumstances the current through the live and neutral conductors will be equal in magnitude but in opposite directions.
- The load-carrying conductors will at any given moment be zero which results in no emf being induced in the secondary windings.
- If, however, there is a fault, an earth-leakage current exists. In this case, the live and neutral conductors do not balance – one conductor carries more current than the other. In this situation, the earth-leakage device produces a change of magnetic flux in the sensing core.
- The emf induced in the secondary winding will activate the trigger circuit. The tripping coil will be energised and the circuit breaker will open and switch the power off. (4)

[10]

TOTAL: 100

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